



Course Name:	Web Development Laboratory (316U01L306)	Semester:	III
Date of Performance:	24/08/2026	DIV/ Batch No:	A1
Student Name:	Nidhish Agarwal	Roll No:	16010125005

Experiment No: 2

Title: Design web Portal using CSS3.

Aim of the Experiment: To apply CSS3 style sheets to an existing HTML5-based Web Portal for the given Use Case study in order to enhance layout, responsiveness, visual aesthetics, and user interaction using modern CSS3 features.

Objectives for the Experiment:

1. Implement different CSS3 styling techniques
2. Design responsive layouts using Flexbox and media queries
3. Enhance user interaction using animations
4. Maintain separation of content and presentation

COs to be achieved:

CO1: Use CSS to prepare the layout of web pages.

Books/ Journals/ Websites references:

1. Students should write

Theory:

Students should write about CSS3 (citation needed for content taken (IEEE or APA7 style))

Problem statement: Demonstrate the implementation of CSS3 features to enhance the layout, responsiveness, and aesthetics of the selected HTML-based web portal for the selected Use Case in experiment 1.

Note: All tasks are compulsory to use.

Task 1: Style the Homepage Header (Inline CSS)

Requirements:

- Use **inline CSS** to:
 - Set background color for header
 - Change text color of title
 - Center align slogan
 - Add padding and border
- Apply inline style to:
 - `<h1>`, `<p>`, `<header>`

Concepts tested:

Inline CSS, color, background, padding, text-align, border.

Task 2: Internal CSS for Navigation Menu

Navigation menu should look professional and consistent across the page.

Requirements:

- Use **internal CSS** (`<style>` in `<head>`)
- Style navigation menu:
 - Remove bullet points
 - Display menu items horizontally
 - Add hover effect
 - Change link colors
- Use:
 - `:hover`, `text-decoration`, `display: inline-block`

Concepts tested:

Internal CSS, selectors, pseudo-class, styling links.

Task 3: External CSS for Whole Website Theme

For a consistent look across all pages.

Requirements:

- Create file: `style.css`
- Link it using `<link>`
- Apply:
 - Body font-family

- Background color
- Common heading color
- Common footer style

Concepts tested:

External CSS, reusable styles, site-wide theming.

Deliverables : style.css linked with book.html

Task 4: Style Categories Sidebar

Categories section should look like a card.

Requirements:

- Use **external or internal CSS**
- Apply:
 - Border
 - Box shadow
 - Padding
 - Rounded corners
- Style list items with:
 - Spacing
 - Hover color change

Concepts tested:

Box model, border-radius, box-shadow, list styling.

Task 5: Card Layout Using CSS Box Model

To make Each item appear as a product card.

Requirements:

- Style a `<div>` as a book card:
 - Width, margin, padding, border
 - Center align content
- Use:
 - `display: inline-block`
 - Add background color and shadow

Concepts tested:

Box model, margin, padding, display property.

Deliverable : link this .css file with book.html

Task 6: Image Gallery Styling with CSS

To make Featured appropriate items/ attributes, whichever applicable in the Use Case look like a grid.

Requirements:

- Use **CSS** to:
 - Set same size for all images
 - Add border and hover zoom effect
 - Add spacing between images
- Use:
 - `transition, transform: scale()`

Concepts tested:

Image styling, hover effects, transitions.

Task 7: Table Styling for Price List**Requirements:**

- Use **CSS** to:
 - Collapse borders
 - Add header background color
 - Zebra-stripping rows
 - Center align text
- Use:
 - `border-collapse, nth-child(even)`

Concepts tested:

Table styling, pseudo-classes.

Deliverable table.css , link to the corresponding file.

Task 8: Style Registration Form**Requirements:**

- **Style:**
 - Input boxes width, padding
 - Submit button color and hover effect

- Fieldset and legend
- Use:
 - :focus, :hover
 - border-radius

Concepts tested:

Form UI styling, pseudo-classes, layout.

Deliverable : form.css

Task 9: Page Layout Using CSS (Header, Sidebar, Content, Footer)

To Create a proper website layout.

Requirements:

- Use CSS to:
 - Set sidebar to left
 - Content to right
 - Footer at bottom
- Use:
 - float or flex
 - Set widths and spacing

Concepts tested:

Layout design, positioning, float/flex.

Deliverable : layout.css

Task 10: CSS Effects & Highlighting.

To attract attention for Special events/data/section, etc..

Requirements:

- Create an “Offer” box:
 - Blinking or glowing effect
 - Different background color
 - Bigger font
- Use:
 - @keyframes animation OR hover effect
 - text-shadow, box-shadow

Concepts tested:

CSS animations, effects, visual emphasis.

Code:

Github link: <https://github.com/nidhishagarwal01/WEB-DEV-LAB-NIDHISH-16010125005>

Output / Screenshot:

Task 1)

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Task 2)

Home Farm Dashboard Irrigation Advisory Sensor Data Contact

Other Pages

- [Farm Information](#)
- [Today's Advisory](#)
- [Photo Gallery](#)
- [Plot Map](#)
- [Irrigation Table](#)
- [Farmer Registration](#)
- [Weather Forecast](#)
- [Awareness Video](#)

About this System

This is a smart irrigation advisory portal made for sugarcane farmers. It uses soil moisture, temperature, humidity and rainfall sensors to suggest when and how much to irrigate. This helps save water and increase crop yield.

Contact

Department of Computer Engineering
K. J. Somaiya School of Engineering, Mumbai-77

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Task 3)

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

[Home](#) [Farm Dashboard](#) [Irrigation Advisory](#) [Sensor Data](#) [Contact](#)

Other Pages

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Task 4)

Other Pages

Farm Information

Today's Advisory

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Plot Map

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Farmer Registration

Weather Forecast

Awareness Video

Task 5)

Sensors Used

Soil Moisture Sensor

Checks water level in
soil.

Temperature Sensor

Checks field
temperature.

Humidity Sensor

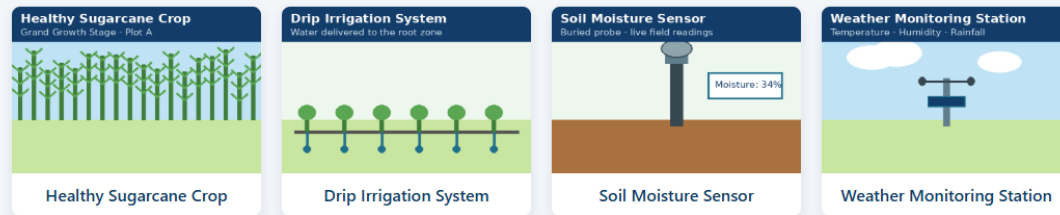
Checks air humidity.

Rainfall Sensor

Detects rainfall in the
field.

Task 6)

Farm Photo Gallery



Task 7)

Irrigation Recommendations for Multiple Plots

Soil Moisture and Irrigation Recommendation - 07 Aug 2026

Plot ID	Farmer Name	Soil Moisture (%)	Weather	Irrigation Recommendation	Water Requirement
PLT-A-07	Ramesh Patil	28	Sunny	Irrigate within 6 hours	45 min (drip)
PLT-B-03	Suresh Jadhav	22	Sunny	Irrigate immediately	60 min (drip)
PLT-B-04	Suresh Jadhav	20	Sunny		65 min (drip)
PLT-C-11	Anita More	41	Cloudy	No irrigation needed	0 min
Average Soil Moisture					27.75%

Task 8)

Farmer Registration Form

Personal Details

Name:

Mobile Number:

Email:

Farm Details

Plot ID:

e.g. PLT-A-07

Village:

Enter village name

Crop Stage:

--Select--

Soil Type:

--Select--

Irrigation Method:

--Select--

Date:

dd - mm - yyyy

Upload Farm Photo:

Choose File

 No file chosen

Register

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Task 9)

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

Home

Farm Dashboard

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Task 10)

Smart Irrigation Advisory System

Smart Irrigation using AI & Sensor Technology

- Home
- Farm Dashboard
- Irrigation Advisory
- Sensor Data
- Contact

Today's Irrigation Recommendation

Water Stress Level: Moderate — Irrigation Needed Soon

Plot ID: PLT-A-07
Farmer Name: Ramesh Patil
Soil Moisture: 28% (measured at 6:00 AM)

Recommended Irrigation Duration: 45 minutes (drip irrigation)
Water Stress Level: Moderate

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Post Lab Subjective/Objective type Questions:

1. Use Bootstrap for CSS.

Answer) Bootstrap is a popular **CSS framework** used to create responsive, attractive, and mobile-friendly websites quickly. It provides many **predefined CSS classes and ready-made components**, such as buttons, navigation bars, forms, cards, tables, alerts, and responsive grids.

Bootstrap reduces the amount of custom CSS code that a developer needs to write. By simply adding Bootstrap classes to HTML elements, we can apply styling, spacing, colors, alignment, and responsive layouts easily.

Conclusion and Discussion:

In this experiment, we learned how to use **CSS to design and style web pages** effectively. We implemented different types of CSS such as **inline, internal, and external CSS** and used properties like colors, borders, spacing, shadows, Flexbox, hover effects, and animations. We also learned how **Bootstrap** can be used to create responsive and attractive web pages using predefined



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classes. Overall, the experiment helped us understand how CSS improves the **appearance, layout, responsiveness, and user experience** of a website.